

INYOUNG KIM

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EDUCATION

University of Illinois Urbana-Champaign

Aug 2024 – Present

Ph.D. in Aerospace Engineering, Expected May 2029

Advisor: Prof. Hiroyasu Tsukamoto

Seoul National University

Mar 2019 – Feb 2024

B.S. in Mechanical Engineering

B.S. in Electrical and Computer Engineering

RESEARCH INTERESTS

Safe and robust autonomy; uncertainty quantification for robotics; nonlinear, adaptive, and robust control; learning-enabled control with provable stability and safety guarantees; foundation models and world models for robotics.

RESEARCH EXPERIENCE

Graduate Research Assistant

Aug 2024 – Present

N-ACXIS Lab, Aerospace Engineering, UIUC

Advisor: Prof. Hiroyasu Tsukamoto

- Built a closed-loop conformal inference framework for nonlinear robust adaptive control using VLM-based parameter predictions.
- Contributing to ongoing research on robust adaptive control for uncertain nonlinear systems, with an emphasis on stability, safety, and robustness guarantees.
- Working on methods for uncertainty quantification and probabilistic guarantees in foundation-model-enabled robotic systems.
- Built and tested aerial-robot experimental platforms for sponsored research projects, including motion-capture integration, controller tuning, flight testing, and data collection.

PUBLICATIONS & MANUSCRIPTS

1. **Inyoung Kim** and Hiroyasu Tsukamoto, “Closed-Loop Conformal Inference with VLM Parameter Prediction for Nonlinear Robust Adaptive Control,” under review at *IEEE Control Systems Letters (L-CSS)*, 2026.

TECHNICAL SKILLS

Programming: Python, MATLAB, C++

Machine Learning / Scientific Computing: PyTorch, NumPy, SciPy, CVXPY

Robotics / Simulation: ROS/ROS2, Genesis(physics simulation)

Computing: Linux, Git, SLURM, GPU computing

CAD: SOLIDWORKS

ADDITIONAL RESEARCH EXPERIENCE

Undergraduate Research Assistant

May 2023 – Aug 2023

Radionavigation Lab, Aerospace Engineering, UT Austin

Advisor: Prof. Todd E. Humphreys

- Debugged a path-generation pipeline for autonomous aerial robots and validated planned trajectories through flight experiments.
- Developed and tested drone behaviors, including music-synchronized flight motions.

Undergraduate Thesis Research

Mar 2022 – Dec 2022

Healthcare Robotics Lab, Mechanical Engineering, Seoul National University

Advisor: Prof. Amy Kyungwon Han

- Developed an MR-compatible actuator for surgical tool manipulation.

Undergraduate Research Assistant

Jul 2021 – Dec 2021

Biorobotics Lab, Mechanical Engineering, Seoul National University

Advisor: Prof. Kyu-Jin Cho

- Worked on a meter-scale wave-energy-harvesting robot.

HONORS & AWARDS

Stillwell Fellowship, UIUC Aerospace Engineering 2024

Beatty Fellowship, UIUC Aerospace Engineering 2024

Student Leadership Award, Seoul National University 2024

Outstanding B.S. Thesis Presentation Award, Seoul National University 2022

Presidential Science Scholarship, Korea Student Aid Foundation 2019 – 2022

Alumni Association Scholarship, Seoul National University Mechanical Engineering 2021

PROFESSIONAL SERVICE

Reviewer, *IEEE Robotics and Automation Letters (RA-L)* 2026

Reviewer, *Distributed Autonomous Robotic Systems (DARS 2026)* 2026

TEACHING EXPERIENCE

Teaching Assistant, University of Illinois Urbana-Champaign Fall 2026

AE353 Aerospace Control Systems

Hold weekly office hours and assist with grading assignments and exams.

Peer Tutor, College Writing 2: Writing in Science & Technology, Seoul National University 7 semesters

LEADERSHIP & OUTREACH

Vice President, Mechanical Engineering Student Council, Seoul National University 2020 – 2021

Mentor of WIA Mentorship Program, Women in Aerospace Engineering, UIUC Fall 2026 – Present